

Chapter 77 — Sexual Function and Dysfunction in the Female

ANATOMY AND PHYSIOLOGY OF FEMALE SEXUAL ORGANS

Introduction

- Anatomy is presented from a **clinical perspective**; **accurate examination of the external and internal genitalia is mandatory** for understanding female sexual function and dysfunction.
- **Physical examination yields critical information** because biological and psychosexual factors are closely interrelated.

Female Sexual Anatomy

- Pelvic organs form a continuum, grouped as **external genitalia (vulva)** and **internal genitalia (vagina, cervix, uterus, fallopian tubes, ovaries)**; **external structures are emphasised** because most physiological research concerns them. The description follows **O'Connell et al. (2008)** and centres on the **distal vagina**.
- **The distal vagina is the site of female sexual response**, interrelated with the clitoris and the urethra; the three **share blood supply and innervation** and **respond as a unit** with the overlying highly vascular vulval skin.
- **Components of the distal vagina**: introitus, vaginal walls, associated exocrine glands.

Vulva

- **Definition**: the wrapping overlying the **clitro-urethro-vaginal complex**.
- **Boundaries**: symphysis pubis anteriorly, anal sphincter posteriorly, ischial tuberosities laterally.
- **Constituents**: mons pubis, clitoris and glands, labia majora and minora, vestibule, hymen, introitus, fourchette, perineal body, urethral orifice, glandular ducts.

Mons Pubis

- **Rounded eminence of subcutaneous adipose connective tissue anterior to the pubic symphysis**; **triangular, apex at the glans clitoris**.
- **Across life**: flat before adolescence → coarse hair with a **horizontal upper boundary** in adolescence → thins after menopause.
- **Fascial support to the clitoris and urethra is provided by the suspensory ligaments of the clitoris**.

Clitoris

- **Erectile structure homologous to the penis**, postero-inferior to the anterior commissure, partially enclosed by the **anterior bifurcated ends of the labia minora**.
- **The corpus clitoridis has two corpora cavernosa** of erectile tissue in dense fibrous tissue, separated by an **incomplete pectiniform septum**; each corpus is **anchored to its ischiopubic ramus by a crus** (the crura are the proximal continuations of the corpora).
- **The deep suspensory ligament (DSL) forms a midline septum separating the left and right dorsal nerves**, which run along the dorsal clitoral body **superficial to, not beneath, the tunica albuginea**.
- **Glans**: small round tubercle with **high cutaneous sensitivity**; histologically **nonerectile vascular tissue**.
- **Suspensory ligament and two ischiocavernosus muscles attach to the crura**.
- **Unlike the penis, the clitoris is separate from the urethra**.

- **Prepuce:** inverted-V skin fold forming a **hood over the glans** and body. **Frenulum:** discrete fold at the posterior glans.

Labia

- **Function:** protect the urethral and vaginal orifices, both opening into the vestibule.

LABIA MAJORA VS LABIA MINORA

Feature	Labia majora	Labia minora
Core tissue	Subcutaneous fat , loose connective tissue, smooth muscle resembling dartos , vessels, nerves, glands	Fat-free spongy tissue , sebaceous and sweat glands, many vessels
Skin	Hair-bearing	Hairless ; pink medial surface continuous with vaginal mucosa, rich in sensory endings
Extent	Mons → perineum	From the clitoris obliquely down, lateral and back ~4 cm, flanking the vaginal orifice
Anterior junction	Anterior labial commissure	Bifurcate; fuse to form the clitoral prepuce
Posterior junction	Posterior labial commissure , over the perineal body; posterior limit of the vulva	Frenulum of the labia minora (may be present in virgins)
Special content	Round ligament of the uterus may terminate here	Dense sensory innervation

- **Gynaecological perineum:** the interval between posterior labial commissure and anus, **2.5–3 cm**.

Vestibule

- **Cavity between the labia minora** containing the vaginal and external urethral orifices, openings of the **two greater vestibular glands** and **numerous mucous lesser vestibular glands**; **vestibular fossa** lies between vaginal orifice and frenulum.
- **Greater vestibular (Bartholin) glands are homologues of the male bulbourethral glands:** small reddish-yellow bodies flanking the vaginal orifice, overlapped by the **posterior vestibular bulb**; each **duct (~2 cm) opens in the groove between hymen and labium minus**.

Introitus

- **Almost closed in childhood** and in some adult pathological states; **highly elastic** to permit the fetal head; **resting diameter ~3–4 cm**.
- **Skin over the introitus and clitoris: hairless, moist, squamous epithelium.**

Hymen Vaginae

- **Thin mucous-membrane fold just within the vaginal orifice;** the orifice appears as a cleft between its folds.
- **Morphology:** usually **annular, widest posteriorly**; semilunar (concave to mons); cribriform or fringed; absent; or **imperforate**. **Carunculae hymenales (myrtiformes)** are remnants after rupture.
- **No established function, yet extreme sociocultural significance; hymenal appearance does not reliably indicate virginal status** given normal variation. The topic interrelates with **female genital mutilation**.

Fourchette

- **Midline skin of the posterior margin of the introitus; important in parturition.**

Perineal Body

- **Pyramidal block of fibromuscular connective tissue between vagina and anal canal;** an **extremely elastic zone at the confluence of the perineal support muscles**, small at rest but stretching without tearing for the fetal head.

Female Urethra and Urethral Orifice

- Detailed anatomy is in the online chapter and in Chapters 101 and 118.

Vulva: Vascularisation, Lymphatic Drainage and Neuroanatomy

VULVAL AND CLITORAL VASCULAR AND LYMPHATIC DRAINAGE

Territory	Arterial	Venous	Lymphatic
Vulval skin	Superficial and deep external pudendal branches of the femoral artery, plus internal pudendal artery, bilaterally	External pudendal → long saphenous vein	Meshwork → 3–4 trunks around the mons → superficial inguinal → deep femoral → pelvic nodes
Clitoris	See clitoral complex	Deep dorsal → internal pudendal vein; superficial dorsal → external pudendal and long saphenous	Deep inguinal; direct efferents to internal iliac
Labia minora	—	—	Deep inguinal
Perineum, lower labia majora	—	—	Rectal lymphatic plexus

- Blood supply is substantial: haemorrhage from vulval injury may be severe.
- Cloquet's node: the last deep femoral node, under the inguinal ligament.

SENSORY INNERVATION OF THE LABIUM MAJUS

Region	Nerve	Segment
Anterior third	Ilioinguinal nerve	L1
Posterior two-thirds	Labial branches of the perineal nerve	S3
Lateral aspect	Perineal branch of posterior cutaneous nerve of thigh	S2

Concept — The vulva straddles a limb-derived anterior-superficial territory (femoral, external pudendal, ilioinguinal) and a pelvic posterior-deep territory (internal pudendal, perineal, sacral); arteries, veins, lymphatics and cutaneous nerves all divide along the same line.

The Clitoral Complex

- Comprises clitoris, distal vagina and urethra, responding as a unit with the overlying vascular skin.
- Female erectile organs are the clitoris and the "vestibular" bulbs; the bulbs are inappropriately named, relating directly to the clitoral components and urethra.
- **Architecture:** clitoral body (corpora) → bilateral crura; separate bulbs posterior to the crura; all unite at the root of the clitoris, covered by the vestibule, related to the distal urethra and meatus, posterior limit the introitus.

ERECTILE VS NONERECTILE VASCULAR TISSUE

	Erectile	Nonerectile
Location	Clitoris (body, crura) and bulbs	Labia minora, glans, periurethral tissue, vaginal walls
Histology	Trabeculated	Vessels dispersed in fibrous matrix, minimal smooth muscle
Arousal	Engorgement with volume expansion	Increased flow without engorgement

- Urethra lies immediately deep to the introital skin, anteriorly in the midline, palpable bulging the anterior distal vaginal wall (epithelium moist, hairless, rugated). Embedded in the anterior vaginal wall on one face, surrounded by erectile tissue (crura) on all others; well shown by magnetic resonance imaging (MRI).

- **Skene (paraurethral) glands** (Skene, 1880): source of urethral secretions at high arousal; immunopositive for prostate-specific antigen (PSA); homologous to the prostate.
- **Female ejaculation** is reported but **lacks firm evidence**.
- **Gräfenberg spot (G-spot)**: anterior vaginal wall over the mid-urethra, sensitive in some women; Gräfenberg located it as arising from the urethra; **anatomical basis controversial**.

Concept — Female erectile anatomy is **one integrated complex wrapped around the distal urethra**; the urethra is vaginal on one face and erectile-encircled on all others, which is why urethral and anterior vaginal wall surgery cannot be separated from sexual function.

Clitoral Complex: Vascularisation and Neuroanatomy

- **Supplied and drained from four directions** (anterolateral and posterolateral bilaterally) plus deep midline drainage.
- **Arteries**: prepuce ← **external pudendal arteries**; erectile tissue ← **dorsal clitoral arteries, perineal arteries** (bulbar and urethral branches), **deep arteries** central in the corpora.
- **Venous**: erectile tissue → **deep dorsal vein** → subpubic canal → **vesical venous plexus**; it **does not accompany the arteries**.
- **Lymphatic**: prepuce → **superficial inguinal**; glans → **deep inguinal directly**. The glans periphery is a **watershed** between external and internal pudendal vessels and between superficial and deep inguinal lymphatics.

SOMATIC VS VISCERAL INNERVATION OF THE CLITORAL COMPLEX

Somatic	Visceral
Pudendal nerve → dorsal nerve of clitoris → skin	Cavernous nerves → arteries of erectile tissue
Afferent limb: sensation	Efferent limb: engorgement

- **The female sexual reflex is unique**: engorgement is its motor component, it can be **voluntarily augmented or suppressed**, and it **pairs somatic afferents with visceral efferents**.
- **Reflex arc**: receptors of clitoral complex and vulva → **somatic afferents** (pudendal: dorsal clitoral and perineal branches) → **S2–S4** → **visceral efferents** (pelvic parasympathetics via **nervi erigentes** through pelvic ganglia) → **effectors** (erectile tissue) with **sub-bulbar and urethral gland secretion**. Visceral afferents from the complex are possible but undemonstrated.
- **Engorgement**: parasympathetic → **dilatation of dorsal and deep clitoral arterioles** → **increased inflow with restricted outflow**; the vulva swells, distal vagina and urethra secrete, and vulval components normally hidden by the labia minora become **exposed and reddened**.

Concept — Clitoral erection is a **mixed reflex: somatic (pudendal) in, parasympathetic (pelvic) out, integrated at S2–S4**; pudendal injury removes the trigger, pelvic or cavernous nerve injury the response.

Figure legends

- **Fig. 77.1** (Kelling et al., 2020): **(A)** frontal view with labia majora and dorsal hood removed — **ascending and descending segments of the clitoral body**, glans, frenulum, crus, vestibular bulb, suspensory ligament, orifices; **(B)** cross-section — superficial dorsal vein, dorsal nerves, deep dorsal artery and vein, deep arteries, corpora within tunica albuginea, clitoral fascia; **the neurovascular bundles lie very superficially on the dorsum**.
- **Fig. 77.3** (Kelling et al., 2020): cadaveric section at the clitoral angle showing the **DSL and both dorsal nerves**; dorsal nerves dissected off the body.
- **Fig. 77.4**: the **DSL inserts at the clitoral angle and forms a midline septum along the body**, separating the dorsal nerves.

- **Fig. 77.5: the suspensory ligament originates at the pubic symphysis; the DSL inserts at the clitoral angle.**
- **Fig. 77.6: internal and external genital organs with MRI correlates; erectile tissue (body, crus, root, bulb), sub-bulbar gland, introitus, perineal body.**

Female Sexual Physiology

- **Urological conditions and procedures can markedly influence female sexual wellness;** sexual function both influences and is influenced by urological disease, and the urologist should address it.

Female Sexual Response

- **Sexual response cycle:** the sequence of physical and emotional changes during arousal and sexually stimulating activity (intercourse, masturbation). **Masters and Johnson (1966): excitement → plateau → orgasm → resolution.**

MASTERS AND JOHNSON PHASES

Phase	Duration	Features
Excitement	Minutes to hours	Clitoral and labia minora swelling, vaginal lubrication and lengthening, nipple erection, genital sensitivity, tachycardia, tachypnoea, pleasure
Plateau	Variable	Intensification; clitoris swollen and highly sensitive; rising respiration, heart rate, blood pressure, muscle tension
Orgasm	Shortest — seconds	Transient peak of pleasure, altered consciousness, rhythmic contraction of pelvic striated circumvaginal muscle with uterine and anal contraction and myotonia; well-being; sex flush
Resolution	Variable	Falling pelvic blood flow, nipple relaxation, resting heart rate, satiety; some women return rapidly to orgasm (multiple orgasm)

- **Female orgasm was denied for centuries.** Many women reach orgasm by **direct or indirect clitoral stimulation**, others with penetration ± clitoral stimulation; **some healthy women never do with penetration;** anal stimulation plays a role for a minority.

FEMALE SEXUAL RESPONSE MODELS

Model	Year	Shape	Stages	Distinguishing idea
Masters and Johnson	1966	Linear	Excitement, plateau, orgasm, resolution	Physiological; no desire phase
Kaplan	1977	Linear	Desire added before arousal, then orgasm	Desire as precursor
Whipple and Brash-McGreer	—	Circular	Seduction → sensations → surrender → reflection	Reflection feeds the next seduction
Basson	2002	Nonlinear / circular	Neutrality → receptivity → arousal → responsive desire → satisfaction	Spontaneous desire not essential; desire may be reactive

- **Linear models fail some women:** some move from arousal to orgasm without desire; others have desire, arousal and satisfaction without orgasm.
- **Orgasm is a cerebral event: sympathetic outflow (T12–L1) → hypogastric nerves → inferior hypogastric plexus → uterovaginal plexus; pudendal reflex contraction** of vaginal, urethral and anal skeletal sphincters; autonomic uterine contraction. **Resolution:** sympathetic arterial constriction and pelvic venous decongestion.

PARASYMPATHETIC VS SYMPATHETIC ROLES

Parasympathetic	Sympathetic
S2–S4 , nervi erigentes	T12–L1 , hypogastric nerves
Arousal : arteriolar dilatation, engorgement	Orgasm and resolution : contraction, then arterial constriction and decongestion

Concept — The models move from **describing physiology** (Masters–Johnson) to **adding motivation** (Kaplan) to **making motivation responsive and relational** (Basson); absent spontaneous desire is therefore not in itself dysfunction.

Figure legend

- **Fig. 77.7: (A)** linear model — arousal against time; **(B)** Basson circular model — **innate desire** and **incentives** → **receptiveness** → **stimuli** → **arousal** → **responsive desire** → **sexual and nonsexual rewards**, feeding back.

Neurogenic Mediators of the Female Sexual Response**CENTRAL CENTRES**

Function	Centre
Desire–arousal integration	Medial amygdala, stria terminalis, ventromedial hypothalamic nucleus, paraventricular nucleus (PVN)
Visual stimuli	Relays to the lateral geniculate nucleus
Arousal, mate selection, genital response	Medial preoptic area
Orgasm	Medial amygdala, anterior cingulate, frontal, parietal and insular cortices, cerebellum

PERIPHERAL MEDIATORS

Mediator	Mechanism	Genital effect	Status
Nitric oxide (NO) from neuronal nitric oxide synthase (nNOS) in cavernous and vaginal nerves	NO → guanylate cyclase → cyclic guanosine monophosphate (cGMP) → calcium sequestration → smooth muscle relaxation	Clitoral erection, vaginal vasodilatation	Core pathway
Arginase	Arginine (NOS substrate) → ornithine	Reduces NO	Inhibitors enhance response in vitro; in vivo data only in male corpus cavernosum
Phosphodiesterase type 5 (PDE5)	Degrades cGMP	Terminates relaxation	PDE5 inhibitors (PDE5Is) promote engorgement; flushing, congestion, headache, visual change; clinical evidence not robust
Vasoactive intestinal polypeptide (VIP)	Nonadrenergic, noncholinergic (NANC); acts with NO	Vaginal blood flow, lubrication, secretion; clitoral erection	Physiological support
Prostaglandins, histamine, tachykinins	Local effectors	Chiefly sensation	Supporting

- **Sequence**: nNOS → NO → guanylate cyclase → cGMP → reduced intracellular calcium → relaxation → engorgement; **PDE5 terminates, arginase starves**.

Hormonal Regulators of Female Sexual Response

Oestrogens

- **Oestradiol is the principal oestrogen; ER- α predominates in genital tissue, ER- β also present.**
- **Maintains genital tissue integrity:** epithelium, stroma, muscularis, rugal thickness, lubrication. **Menopause** reduces genital sensitivity, vaginal thickness, collagen, moisture and **acidity (pH rises)**, possibly via **VIP downregulation**.
- **Neural:** oestradiol **expands pudendal touch-receptor fields** (animal), lowering sensory thresholds. **Vascular:** **vasoprotective and vasodilatory**, protecting the **hypogastric-pudendal arterial bed** from atherosclerosis.
- **NO machinery:** oestrogen loss (ageing, surgical castration) → **reduced vaginal NO and NOS, wall fibrosis, epithelial and smooth muscle apoptosis**; replacement reverses all three.
- **Deficiency** → thin, fragile, dry, less acidic vagina → **dryness and dyspareunia**. Serum oestradiol ≤ 50 pg/mL markedly increases risk of dryness and sexual pain.
- **Vaginal or systemic replacement** → more interest and enjoyment, less pain, greater orgasmic potential. Menopausal urogenital symptoms (frequency, urgency, dysuria, incontinence) **may be exacerbated by intercourse**.

Testosterone

- **Low testosterone is associated with reduced arousal, libido, responsiveness, genital sensation and orgasm.** Premenopausal production is roughly equal from adrenals and ovaries.
- **Sex hormone-binding globulin (SHBG)-bound testosterone is biologically inactive;** albumin binding is weak. **Androgen receptors (ARs)** are present in genital tissue; AR gene CAG-repeat length may modulate response (scant evidence).
- **Efficacy:** **parenteral oestrogen-androgen combinations outperform oestrogen alone** for desire, libido, energy, well-being, arousal, fantasies, enjoyment and orgasm; testosterone with oestrogen treats **dyspareunia, vulvovaginal atrophy and poor lubrication**; topical testosterone has been used in vulval lichen disease.

VIRILISATION WITH ANDROGEN THERAPY

Early, reversible	Late, largely irreversible
Acne, hirsutism, menstrual irregularity	Male-pattern baldness, worsening hirsutism, voice change, clitoral hypertrophy

- **Cardiovascular safety:** long-term data needed; a long-term longitudinal study found **no link between sex hormones and cardiovascular mortality**; **meta-analysis: no adverse cardiovascular effect in hypoactive sexual desire disorder (HSDD)**; no increased cardiovascular disease in transgender men on androgens. **Benefit in HSDD is established; long-term safety studies are still needed.**

OESTROGEN VS TESTOSTERONE

	Oestrogen	Testosterone
Role	Structural and vascular maintenance	Desire and responsiveness
Deficiency	Dryness, dyspareunia, atrophy, fibrosis	Low libido, arousal, sensation, orgasm
Molecular link	Regulates NOS and VIP	SHBG governs bioavailability; genital AR

Oestrogen	Testosterone
Principal risk —	Virilisation ; cardiovascular risk debated, not substantiated

Concept — *Oestrogen maintains the substrate (epithelium, vessels, NO machinery), so its loss produces structural complaints; testosterone drives the motivational layer, so it is the agent for HSDD, traded against virilisation.*

Mental Aspects of Sexual Response

- **Psychosocial variables interact significantly with sexual function:** mood, trauma, relational factors, personality; also medical and mental illness, medications, family and cultural beliefs, early experiences, partners, stressors. **Exploring them reveals modifiable obstacles.**
- **Depression:** substantial concurrent depression among women with arousal disorders. **Anxiety is bidirectional** — anxiety disorders worsen function, yet a minority report **increased interest when anxious.**
- **Stress:** daily stressors → more problems, less satisfaction; **chronic stress lowers laboratory genital response.** **Childhood and adolescent sexual abuse** is significantly associated with adult dysfunction.
- **Relational factors:** conflict, longer duration, dissatisfaction, desire or script discrepancy, lack of intimacy or trust, incompatibility, poor stimulation, non-optimal context, **partner's dysfunction or poor technique**; data mainly **self-reported and cross-sectional.**
- **Conservative gender stereotypes** (woman as submissive recipient) impair arousal and orgasm; fading but prevalent in some societies.
- **Assess need for psychotherapy:** stressors, social situation, relationships, psychiatric history, **sexual trauma.**

Concept — *Female sexual function is **biopsychosocial**, and the psychosocial arm is frequently the modifiable one; because the same variable (anxiety) can inhibit or facilitate, **causation must be established in the individual, not assumed.***

EVALUATION OF SEXUAL WELLNESS

History Taking

- **Patients are reluctant, so the clinician should raise sexual health;** only a small minority of women with sexual problems seek advice.
- **Barriers:** inadequate training and confidence, perceived lack of treatments, lack of time, patient hesitancy, **underestimated prevalence.**
- **Use a broad, sex-positive, health-focused framework;** sexual health is **a right for all** (American College of Obstetricians and Gynecologists (ACOG) Committee Opinion No. 706, 2017).
- **LOFTI** (listening, observing, feelings, thinking, interpreting) for **psychosexual problems.**
- **Assessment integrates biological, psychological, sociocultural and interpersonal factors.**

TABLE 77.1 BIOPSYCHOSOCIAL MODEL FOR ASSESSING SEXUAL (DYS)FUNCTION (MODIFIED FROM BITZER ET AL., 2013; FUGL-MEYER ET AL., 2013; LATIF AND DIAMOND, 2013)

Domain	Factors
Biological	Medications, hormonal status, neurobiology, physical health, ageing
Psychological	Depression, anxiety, self-image, substance abuse, sexual abuse or trauma
Sociocultural	Upbringing, cultural norms and expectations, religion

Domain	Factors
Interpersonal	Relationship status and quality, partner's sexual function, life stressors

- **Purpose:** ensure no domain is skipped. **Pattern:** each domain maps onto the physiology (hormones, mental aspects, culture, relationship). **Trap:** interpersonal is a separate fourth domain — partner's function belongs there.
- **Counsel on the wide range of normal:** contrast the linear Masters–Johnson model with **Basson's 2002 Sexual Response Circle**, which incorporates emotional intimacy and satisfaction; **discuss intimacy and relationship satisfaction, not only stimuli.**

PLISSIT Model of Assessment and Treatment

- **Annon (1976);** efficient in primary care and **embeds escalation: Permission → Limited Information → Specific Suggestions → Intensive Therapy.**

TABLE 77.2 PLISSIT MODEL OF ASSESSMENT (MODIFIED FROM FAUBION AND RULLO, 2015)

Step	Action	Content	Illustrative phrasing
Permission	Permit discussion; normalise	Validation	Thanks the patient; lower postmenopausal desire is commonly reported
Limited information	Basic sex education	Response cycle, ageing, anatomy; foreplay; contributing conditions and medications	Responsive may replace spontaneous desire after menopause
Specific suggestions	Simple measures	Lubricants, vaginal oestrogen, position change, vibrators, intimacy	Plan sexual activity more intentionally
Intensive therapy	Validate and refer	Sex therapist, couples counsellor, subspecialist	Needs specialist expertise

- **Purpose:** stepped escalation — most needs met low down. **Trap:** intensive therapy means referral; vaginal oestrogen is a specific suggestion.
- **Open by normalising;** screening questions: permission, current activity, **partners male, female or both,** concerns. **Screen all ages** by questionnaire or review of systems.
- **Brief Sexual Symptom Checklist for Women (BSSC-W):** from the **International Consultation in Sexual Medicine (ICSM);** ACOG-recommended but not validated.

TABLE 77.3 BRIEF SEXUAL SYMPTOM CHECKLIST FOR WOMEN (BSSC-W)

Item	Content	Format
1	Satisfied with sexual function?	Yes/No — stop if Yes
2	Duration of dissatisfaction	Free text
3	Problems: interest; genital sensation; lubrication; orgasm; pain; other	Multiple choice
4	Most bothersome problem	Select one
5	Wishes to discuss with doctor?	Yes/No

- **Purpose:** triage; item 3 maps onto **desire, arousal (sensation, lubrication), orgasm, pain.** **Trap:** ACOG-recommended ≠ validated.
- **Open-ended, gender-neutral routine questions** (activity, satisfaction, concerns, partner satisfaction, orgasm, frequency, **lubrication**) help **discriminate FSD types.**
- **Comprehensive history:** **precise concern,** associated symptoms, **chronology,** triggers, modifying factors, **previous treatments.**

TABLE 77.4 ACOG COMMITTEE RECOMMENDATIONS ABOUT COMPREHENSIVE SEXUAL HISTORY TAKING (COMMITTEE OPINION NO. 706, 2017)

Domain	Core enquiries
Partners	Current or past activity; partner numbers (recent months, 12 months); male, female or both — if both, repeat for each
Practices	Explicit, framed on sexually transmitted infection (STI) risk: genital, anal, oral contact (12 months)
Protection from STIs	Use, type, frequency ; reasons for non-use; comfort requesting condoms; intermittent-use circumstances
History of STIs	Previous STI , treatment, recurrence; human immunodeficiency virus (HIV)/STI testing and wish to test; partner's STI history
Prevention of pregnancy	Conception intent or concern; contraception ; partner support; information needs
Completion	Open questions on other concerns

- **Pattern:** who → what → risk mitigation → consequences; mnemonic **five Ps** (partners, practices, protection, past STIs, pregnancy prevention). **Trap:** partner gender is asked, not assumed.
- Refer if untrained for in-depth discussion.

Questionnaires

- Detailed discussion online; Table 77.5 printed.

TABLE 77.5 SUMMARY OF SELECTED QUESTIONNAIRES/SCALES USED TO EVALUATE FEMALE SEXUAL (DYS)FUNCTION (MODIFIED FROM HATZICHRISTOU ET AL., 2016)

Purpose	Instrument	Items	Population	Domains	Distinguishing feature
Global	Female Sexual Function Index (FSFI)	19; 6	Pre/postmenopausal; hetero- and homosexual	Desire, arousal, lubrication, orgasm, satisfaction, pain	Most studied and used ; multilingual; 6-item screen ; cancer and chronic pelvic pain versions
	Sexual Function Questionnaire (SFQ)	28	Heterosexual, active in 4 weeks	Desire, arousal, orgasm, pain, enjoyment, partner	Reliable; discriminates arousal disorder and HSDD
	Brief Index of Sexual Functioning for Women (BISF-W)	22	Heterosexual	Desire, arousal, frequency, receptivity, orgasm, relationship, problems	Easy; moderate reliability
	Arizona Sexual Experience Scale (ASEX)	5	Heterosexual	Desire, arousal, lubrication, orgasm	Both sexes ; screening
	Golombok Rust Inventory of Sexual Satisfaction (GRISS)	28	Heterosexual	Avoidance, nonsensuality, infrequency, vaginismus , anorgasmia , noncommunication, dissatisfaction	Both sexes ; compares treatments
Desire	Profile of Female Sexual Function and Personal Distress Scale (PFSF)	37	Postmenopausal, low desire	Desire, arousal, orgasm, pleasure, concerns, responsiveness, self-image	Oophorectomised women ; multicentre trials

Purpose	Instrument	Items	Population	Domains	Distinguishing feature
	Sexual Interest and Desire Inventory (SIDI)	13	Premenopausal, low desire	HSDD	HSDD severity and response
Distress	Female Sexual Distress Scale–Revised (FSDS-R)	13; 1	Dissatisfied women	Relationship-based distress	Single item screens midlife distress
Condition	Pelvic Organ Prolapse/Urinary Incontinence Sexual Questionnaire (PISQ)	31; 12	Prolapse, urinary and faecal incontinence	Behavioural–emotive, physical, partner	Only validated instrument for prolapse and incontinence; International Urogynecological Association (IUGA) revision PISQ-IR
Body image	Female Genital Self-Image Scale (FGSIS)	7	Women	Genital self-image	US nationally representative validation

- **Purpose:** choose by question — global, desire, distress, condition, body image. **Pattern:** mostly heterosexual validation; **FSFI includes homosexual women; ASEX and GRISS both sexes;** HSDD tools split by menopause (PFSF post, SIDI pre). **Trap:** **PISQ is the urogynaecological instrument; FSFI measures function, FSDS-R distress.**

Comorbidities, medications and gynaecological history

- **History must cover comorbidities, medications and surgery** affecting neurological, endocrine, vascular or psychological function. **Menopause is not proven to cause universal decline.**
- **Chronic conditions contributing to FSD:** diabetes mellitus (DM), hypertension, overactive bladder (OAB), multiple sclerosis (MS), spinal cord injury (SCI), **major depressive disorder.**

TABLE 77.6 MEDICAL CONDITIONS THAT CAN AFFECT FEMALE SEXUAL FUNCTION (MODIFIED FROM FAUBION AND RULLO, 2015)

Impact	Conditions
Low desire	DM, hypertension, Parkinson disease, dementia
Arousal via pelvic perfusion	Coronary artery disease
Lubrication and orgasm	Hypothyroidism
Genital pain and lubrication	Dermatoses: lichen sclerosus, lichen planus, eczema
Desire, arousal, pain	Urinary incontinence
All domains	Malignancy and treatment (breast, anal, colorectal, bladder, gynaecological); neuromuscular disorders, SCI, MS

- **Pattern:** metabolic disease → desire; vascular → arousal; skin → pain; **neurological disease and cancer → all domains.** **Trap:** DM and hypertension sit under **low desire**, not arousal.

MEDICATIONS AND MECHANISMS

Mechanism	Drugs	Effect
Central mood and libido	Antidepressants, antipsychotics, sedatives	Low desire
Reduced genital flow	Certain antihypertensives, anti-oestrogens	Low arousal, lubrication

Mechanism	Drugs	Effect
Raised SHBG, lower free testosterone	Oral contraceptives (OCs)	Low desire — only a small minority affected

- **Illicit drugs and alcoholism** are associated with FSD; **heavy smoking** causes genital vascular insufficiency.
- **Urogynaecological and obstetric causes:** endometriosis, fibroids, infection, **pelvic organ prolapse (POP)**, hysterectomy; **difficult delivery, episiotomy, vaginal surgery** (denervation, dyspareunia); **surgical castration, worst when premenopausal.**

Evaluation of the Partner

- **Partner involvement is of utmost importance;** partners of women with FSD often have problems of their own. **Treating the male partner's erectile dysfunction (ED) improves the woman's sexual life.** Consider partner treatment or referral.

Physical Examination

- **A focused internal and external pelvic examination identifies causes,** especially in dyspareunia and vaginismus. Common findings: **vaginal atrophy**, dryness, infection, **vulval dermatoses**, **pelvic floor muscle (PFM) dysfunction**, adnexal masses, deep pelvic pain. **A normal examination is reassuring.**
- **External:** inspect mons → labia majora → labia minora → clitoris → vestibule **gently with a gloved hand and cotton swab.** **Vestibular redness and pain → vestibulitis; flattened pale labia → oestrogen deficiency or POP;** also hair distribution, labial redundancy, atrophy, lesions.
- **Internal:** palpate PFMs, urethra, anus; **bimanual** for uterine size, flexion, adnexal masses; **fornix tenderness → endometriosis;** POP and **urethral hypermobility at rest and straining.** **Normal PFMs contract and relax voluntarily and are not tender.**
- **Neurological:** lower-limb sensory and motor; **lumbosacral screen** — PFM strength, **anal resting tone, voluntary anal contraction, perineal sensation;** genital neuropathy by **heat/cold, vibration, toothpick or pin;** abnormal screen → full examination.
- **Continence:** lower urinary tract symptoms (LUTS) associate with sexual problems, so perform **cough or Valsalva stress test and Q-tip test.**

Pelvic floor muscles

- **Sexual function depends on normal PFMs** (Kegel 1952); **pubococcygeus and iliococcygeus produce orgasmic contractions;** stronger contraction → **higher FSFI orgasm and arousal scores.**

PFM CATEGORIES

Category	Behaviour	Consequence
Normal	Voluntary and involuntary contraction and relaxation	—
Overactive (high tone)	Fails to relax; may contract during voiding or defaecation	Voiding and defaecatory dysfunction, dyspareunia
Underactive (low tone)	Cannot contract voluntarily	Urinary/faecal incontinence, POP
Nonfunctioning	Completely inactive	—

- **Digital rectal or vaginal palpation is commonest** (tone, squeeze, symmetry, relaxation; maximal = strength, sustained = endurance); invasive alternatives are manometry and dynamometry. **No validated strength scale;** tenderness graded by **Lukban and Whitmore (2002):** 0 none, 1 comfortable pressure, 2

uncomfortable pressure, 3 moderate pain intensifying with contraction, **4 severe pain preventing contraction.**

Vulvodynia

LOCALISED VS GENERALISED VULVODYNIA

Localised (provoked)	Generalised (unprovoked)
Pain provoked by light touch of a moistened cotton swab: vestibule, interlabial sulci, introitus, peri-clitoral	Unprovoked stinging, burning, rawness or pain anywhere on the vulva , unexplained
Diagnosed by the Q-tip test	Examination often normal , or tenderness, hyperaesthesia, hypoaesthesia
Lidocaine 5% ointment or EMLA (lidocaine–prilocaine) 10 minutes before intercourse	—

Laboratory Tests

- **Rarely indicated** without a specific suspected condition; baseline **serum chemistry, lipids, glycated haemoglobin (HbA1c)** may reveal DM.
- **Testosterone: no female normal range, no correlation with libido — random measurement is worthless**; measure only if a **hyper- or hypoandrogenic state** is suspected.
- **Suspected hormonal abnormality**: screen for **prolactinoma, thyroid and adrenal disorders**; **dehydroepiandrosterone sulphate (DHEA-S)** for adrenal insufficiency, being **solely adrenal**.
- **Androgen insufficiency syndrome** (free testosterone **below lowest quartile**): low interest and well-being, fatigue, **vasomotor symptoms persisting despite oestrogen**, low motivation.
- **Routine oestrogen measurement is not useful**; progesterone has very limited value.

Physiological Measures of Sexual Function

- Detailed discussion online. **Mostly invasive, poorly defined, lacking standardisation, validity and reliability — research tools.**

TABLE 77.7 METHODS TO ASSESS VULVOVAGINAL BLOOD FLOW (MODIFIED FROM LEVIN ET AL., 2016)

Site	Method	Signal	Principle
Vagina	Oxygen perfusion and heat dissipation (Levin, 2006)	O ₂ diffusion, heat clearance	Heated suction-cup electrode
	Vaginal photoplethysmography (Sintchak et al., 1975)	Back-scattered light	Tampon-sized infrared probe; back-scatter $\propto$ wall blood volume
Labia	Labia minora oxygen perfusion (Sommer et al., 2001)	Perfused O ₂	Heated electrode on adhesive ring
	Labial photoplethysmography (Prause et al., 2005)	Back-scattered light	Clip on labia majora
	Thermistor clip (Henson et al., 1978)	Temperature	Clip on labia minora ; temperature as flow proxy
Clitoris + vagina	Combined photoplethysmography (Gerritsen et al., 2009)	Back-scattered light	Clitoral sensor on vaginal probe
Imaging	Thermal imaging (Kukkonen et al., 2007)	Infrared (electromagnetic) emission	Temperature as flow proxy
	Doppler ultrasonography (Garcia et al., 2005)	Ultrasound echoes	Imaged echoes
		Light frequency shift	Superficial skin flow

Site	Method	Signal	Principle
	Laser Doppler imaging (Waxman et al., 2009)		
	MRI (Maravilla et al., 2008)	Signal intensity	Engorgement = increased signal

- **Pattern:** every method measures a **proxy**; temperature methods rest on an assumption. **Trap: none is validated for clinical diagnosis.**

Figure legends

- **Fig. 77.8** (Apgar et al., 2002): **atrophic vulva** — pale thin vestibule; **loss of labial fullness, minimal moisture, pallor of urethra or vagina, narrow introitus, loss of rugae.** pH paper on the vaginal wall: **oestrogenised pH 3.5–4.5; pH $\geq$ 4.5 without infection or semen indicates atrophy.**
- **Figs. 77.9–77.10:** vulval dermatoses —

LICHEN SCLEROSUS VS LICHEN PLANUS

	Lichen sclerosus (Fig. 77.9)	Lichen planus (Fig. 77.10)
Nature	Chronic progressive inflammatory dermatosis, unknown aetiology	Chronic desquamative erosive dermatitis
Signature	Plaques, oedema, loss of architecture , introital narrowing	Wickham striae — the only pathognomonic sign
Distribution	Labia; extends to perineum and perianal skin	Labia minora, vestibule; anus rarely
Age	—	50–60 years
Symptoms	Late dyspareunia from stenosis, fissuring, agglutination	Severe pruritus , pain, burning
End-stage	Introital stenosis	Tissue destruction, vaginal stenosis

- **Fig. 77.11** (Shah and Hoffstetter, 2014): **cotton-swab mapping** clockwise — 1–2 inner thighs, 3–5 labia majora, 6–8 interlabial sulci, 9 clitoris and hood, 10 perineum, **11 vestibule.**

KEY POINTS: EVALUATION OF SEXUAL WELLNESS

- **Clinicians should raise sexual health;** women are usually reluctant to.
- **BSSC-W screens; FSFI supports a comprehensive history.**
- **Thorough physical examination identifies causes.**
- **Laboratory evaluation is rarely indicated** without a specific concern.
- **Physiological tests lack standardisation, validity and reliability.**

Concept — Evaluation is a **filter**: normalise and screen broadly, characterise with a structured history, a purpose-matched instrument and an examination that **sees and feels** the informative findings, keep laboratory testing sparing, and **escalate to specialist therapy only when lower tiers fail.**

DEFINITIONS OF SEXUAL DYSFUNCTIONS IN FEMALES

- **Most widely used systems:** International Classification of Diseases, 10th Edition (ICD-10) of the World Health Organization, and the Diagnostic and Statistical Manual of Mental Disorders (DSM-IV-TR, 2000; DSM-5, 2013) of the American Psychiatric Association; **DSM-5 and ICD-10 are the two officially sanctioned systems.**

ICD-10 VS DSM-5

	ICD-10	DSM-5
Orientation	Medical	Psychiatric

	ICD-10	DSM-5
Structure	Organic vs nonorganic	Criteria-based disorders
Organic codes	Vaginismus and dyspareunia of organic aetiology	—
Nonorganic codes	Lack of desire; aversion or lack of enjoyment; failure of genital response ; orgasmic dysfunction; nonorganic vaginismus and dyspareunia ; excessive sexual drive ; two nonspecific codes	Table 77.8

DSM-5 Definitions of Sexual Dysfunctions in Females

- Detailed discussion online.

TABLE 77.8 DSM-IV-TR AND DSM-5 DEFINITIONS OF FEMALE SEXUAL DYSFUNCTION

Domain	DSM-IV-TR	DSM-5
Desire	HSDD : deficient or absent fantasies and desire. Sexual aversion disorder : aversion to and avoidance of genital contact	Merged → female sexual interest/arousal disorder (FSIAD) , ≥3 of 6: (1) reduced interest; (2) reduced thoughts or fantasies; (3) reduced initiation, unreceptive; (4) reduced excitement or pleasure in 75–100% of encounters; (5) reduced response to erotic cues; (6) reduced genital or nongenital sensation in 75–100% . Aversion not retained
Arousal	Female sexual arousal disorder (FSAD) : inability to attain or maintain an adequate lubrication–swelling response	Absorbed into FSIAD
Orgasm	Female orgasmic disorder (FOD) : delay or absence after normal excitement	FOD : in 75–100% — marked delay, infrequency or absence, or markedly reduced intensity
Pain	Dyspareunia ; vaginismus — involuntary perineal muscle contraction around the outer third of the vagina	Merged → genito-pelvic pain/penetration disorder (GPPPD) , ≥1 of: penetration difficulty; marked pain; marked fear or anxiety about pain; marked PFM tensing

- **Pattern**: DSM-5 collapsed six female diagnoses into three — **desire + arousal merged**, **dyspareunia + vaginismus merged**, **aversion dropped**, **orgasm quantified with intensity added**. **Trap**: DSM-5 has no separate female HSDD, FSAD or vaginismus.

ICSM DEFINITIONS OF SEXUAL DYSFUNCTIONS IN FEMALES

- **ICSM**: large international expert panel; **aetiology-independent terminology**; **consultation 2015**, **female definitions 2016** — a **synthesis of DSM-5, DSM-IV-TR and ICD-10** plus new consensus definitions. Some are **immature**, many **expert opinion only**, so **no reliable grade of recommendation**.

ICSM DEFINITIONS

Entity	Definition
Hypoactive Sexual Desire Disorder	Persistent or recurrent deficiency or absence of sexual or erotic thoughts or fantasies and desire for sexual activity
Female Sexual Arousal Disorder	Persistent or recurrent inability to attain or maintain arousal until completion, on adequate subjective assessment of the genital response
Persistent Genital Arousal Disorder (PGAD)	Spontaneous, intrusive, unwanted genital arousal without interest or desire; usually unpleasant; not relieved by at least one orgasm ; lasts hours or days
Female Genital-Pelvic Pain Dysfunction	≥1 of: pain on penetration ; marked pain during genital contact ; marked fear or anxiety about such pain; marked PFM hypertonicity or overactivity
Female Orgasm Disorders	

Entity	Definition
	Distressing compromise of orgasm frequency, intensity, timing or pleasure: anorgasmia, reduced frequency, muted intensity, premature or delayed , anhedonic
Female Orgasmic Illness Syndrome	Negative feelings or physical symptoms after sexual activity — headache, malaise, fatigue

- **International Society for the Study of Women's Sexual Health (ISSWSH)** subdivision of FSAD:

FEMALE COGNITIVE VS GENITAL AROUSAL DISORDER

Female cognitive arousal disorder (FCAD)	Female genital arousal disorder (FGAD)
≥6 months, distressing difficulty with mental excitement — not feeling "turned on"	≥6 months, distressing difficulty with genital response to adequate stimulation
May coexist with FGAD or HSDD	Not diagnosed if stimulation is not aligned with her preferences
Behavioural and psychotherapeutic treatment	Pelvic anatomy and physiology — within urological scope

DEFINITION FRAMEWORKS COMPARED

	ICD-10	DSM-5	ICSM 2016	ISSWSH
Basis	Medical	Psychiatric	Expert synthesis , aetiology-independent	Expert nomenclature
Desire–arousal	Separate	Merged (FSIAD)	Separate	Separate; FSAD → FCAD + FGAD
Pain	Organic vs nonorganic	Merged (GPPPD)	Genital-pelvic pain dysfunction	—
Extra entities	Excessive drive	—	PGAD, orgasmic illness syndrome	—
Status	Official	Official	Partly expert opinion, ungraded	Consensus

Concept — *The systems disagree chiefly on **whether desire and arousal are separable**: DSM-5 merges them on nonlinear-response grounds, ICSM and ISSWSH keep them apart and split arousal further — and the FGAD split is what brings arousal disorder into urological practice.*

EPIDEMIOLOGY OF FEMALE SEXUAL DYSFUNCTION

- Detailed discussion online.

KEY POINTS: DEFINITIONS AND EPIDEMIOLOGY

- ICD-10 (WHO), DSM-IV-TR and DSM-5 are the most widely used systems; their terminology differs.
- The ICSM consensus is an expert-approved synthesis of ICD and DSM.
- Epidemiological literature on FSD is markedly heterogeneous in quality.
- HSDD is the most prevalent FSD.

FEMALE SEXUAL DYSFUNCTION IN SPECIAL POPULATIONS

LGBTQ+

- **Transgender and gender-diverse people are increasingly visible; experienced gender is fundamental to the self.** Needs: **gender-affirming hormones**, surgery, **fertility services**, routine urological and urogynaecological care.

- **Language and pronouns are critical**; gendered language contributes to disparities; **simple adjustments in framing measurably improve care**. Detailed discussion online and in Chapter 126.

TABLE 77.9 BASIC TERMINOLOGY RELATED TO TRANSGENDER PEOPLE

Term	Meaning
Gender identity	One's own sense of gender (male, female, genderqueer); not restricted to anatomy or social role
Nonbinary	Neither male nor female; outside rigid binary definitions
Queer	Flexible alternative to "gay" or "lesbian"
MSM / WSW	Men who have sex with men / women who have sex with women — behavioural terms
Top surgery	Breast reduction or reconstruction
Bottom surgery	Genital gender-affirming surgery

CIS AND TRANS TERMS

	Identifies as female	Identifies as male
Assigned female at birth	Cis female	Trans male
Assigned male at birth	Trans female	Cis male

- **Pattern**: cis/trans = concordance of assigned sex and identity; the noun names the **identity**. **Trap**: a **trans male was assigned female at birth**; MSM and WSW describe behaviour.

TABLE 77.10 MOST COMMON CANCER TYPES AFFECTING LGBTQ+ POPULATION

Sites
Anal, breast, cervical, colorectal, endometrial, lung, prostate

- **Pattern**: organ-based — **screening follows the organs present**. **Trap**: **anal cancer** is the site absent from routine general screening.

TABLE 77.11 RISKS ASSOCIATED WITH SEX HORMONE TREATMENT

Hormone	Very high risk	Moderate risk
Oestrogen	Thromboembolic disease	Cerebrovascular and coronary disease, breast cancer , hypertriglyceridaemia , cholelithiasis
Testosterone	Erythrocytosis (haematocrit >50%)	Severe liver dysfunction , coronary and cerebrovascular disease, hypertension, breast or uterine cancer

- **Pattern**: one signature very-high risk each, **opposite haematological problems** (clot vs red cell mass); vascular risks shared. **Trap**: testosterone carries moderate **breast and uterine cancer** risk — retained organs remain at risk.

TABLES 77.12 AND 77.13 RECOMMENDED FOLLOW-UP FOR TRANS MALE AND TRANS FEMALE PATIENTS

	Trans male (testosterone)	Trans female (oestrogen ± antiandrogen)
Review	3-monthly in year 1, then 1–2 times yearly (virilisation, adverse effects)	—
Hormones	Testosterone 3-monthly until normal male range	Testosterone and oestradiol 3-monthly ; testosterone <50 ng/dL; oestradiol not above physiological peak 100–200 pg/mL
Safety bloods	Haematocrit or haemoglobin at baseline, 3-monthly in year 1, then 1–2 times yearly	Spironolactone: potassium 3-monthly in year 1, then yearly
Cardiometabolic	Weight, blood pressure, lipids	—
	—	As for non-transgender individuals

	Trans male (testosterone)	Trans female (oestrogen ± antiandrogen)
Cancer screening		

- **Pattern:** quarterly in year 1, then yearly; each schedule monitors its regimen's signature risk. **Trap:** in trans females testosterone is **suppressed** and oestradiol **capped**.

Sexuality and Disability

- Detailed discussion online.

KEY POINTS: SPECIAL POPULATIONS

- Ask sexual orientation and gender identity in every history to identify disparities and problems.
- Screen and monitor transgender patients for gender-affirming hormone side effects.
- Discuss fertility preservation before medical or surgical transition — effects may be irreversible.
- Evaluate sexual dysfunction in disability within the context of disability and comorbidity, starting with attitudes to sexuality and relationships.
- Women with SCI commonly have impaired genital sensation, lubrication and orgasm, with chronic genital pain and spasticity.
- Sexual health in SCI must not be overlooked.

FEMALE SEXUAL INTEREST DISORDER

- About 40% of women experience sexual dysfunction in their lifetime; lack of desire is the commonest at all ages (HSDD in DSM-IV-TR; within FSIAD in DSM-5). Distressing low desire affects a minority.
- With ageing, desire falls but so does the distress it causes — low desire nearly universal in older women, HSDD uncommon.
- Distressing low desire is associated with lower health-related quality of life (HRQoL), dissatisfaction with sex life or partner, frustration, hopelessness, anger, low self-esteem, **loss of femininity**; mental-health decrements; in postmenopausal women more fatigue, depression, memory problems, back pain and lower quality of life.

Definition

HSDD DEFINITIONS COMPARED

	ICD-11 (WHO)	ISSWSH (Parish et al., 2016)	DSM-IV-TR	DSM-5
Name	Hypoactive sexual desire dysfunction	HSDD	HSDD	FSIAD (merged)
Core	Absent or reduced desire or motivation: reduced spontaneous desire; reduced responsive desire; inability to sustain desire	Lack of motivation (spontaneous or responsive) and/or loss of desire to initiate or participate, including avoidance	Deficient or absent fantasies and desire	≥3 of 6 criteria
Duration	Several months	≥6 months	Persistent or recurrent	≥6 months
Distress	Required	Required — frustration, grief, guilt, incompetence, loss, sadness, sorrow, worry	Marked distress or interpersonal difficulty	Personal, not partner-driven
Exclusions	Not secondary to sexual pain		Not due to medical or psychiatric condition,	—

ICD-11 (WHO)		ISSWSH (Parish et al., 2016)	DSM-IV-TR	DSM-5
		Not secondary to pain, abuse, partner dissatisfaction or life stressors	substance or medication	
Responsive desire	Yes	Yes	No	Implicit (merger rationale)

- Exclude relationship conflict as the primary cause before diagnosing HSDD. Subtypes: lifelong or acquired; generalised or situational.
- DSM-5 merger rationale: response is **not always linear**; the desire–arousal distinction **may be artificial**.

Pathophysiology

CONTRIBUTORS TO LOW DESIRE

Domain	Contributors
Disease and drugs	Hypertension, hypothyroidism, DM; calcium channel blockers, angiotensin-converting enzyme (ACE) inhibitors
Reproductive states	Pregnancy, breastfeeding, postmenopause
Ageing	Middle-aged women have the highest prevalence of distressing low desire; declining testosterone and oestrogen; diminished genital sensation needs more intense, longer stimulation; hypo-oestrogenism → atrophy → dyspareunia → low desire
Psychiatric	Depression, anxiety; selective serotonin reuptake inhibitors (SSRIs), anxiolytics
Psychological	Childhood or pubertal abuse, stress, distraction, self-focused anxiety, personality disorder, body image
Sociocultural	Highly conservative cultural or religious upbringing
Relational and stress	Conflict; partner's ED or premature ejaculation; financial, career, family stressors

- Hormone levels do not define HSDD: no testosterone level predicts HSDD; low oestradiol associates with low desire but **not all women with HSDD have low oestradiol**.
- Desire balances excitatory and inhibitory neuromodulators:

EXCITATORY AND INHIBITORY MODULATORS OF DESIRE

Source	Excitatory	Inhibitory
Text	Dopamine, noradrenaline, melanocortins, oxytocin, vasopressin	Serotonin, opioids, endocannabinoids, prolactin
Fig. 77.12 (Clayton, 2010)	Dopamine, melanocortin, testosterone, oestrogen	Serotonin, opioids, prolactin, progesterone

- The figure adds the **sex steroids** (testosterone and oestrogen excitatory, progesterone inhibitory).
- **Imaging in HSDD: reduced activity in arousal regions** (medial orbitofrontal cortex, periaqueductal grey) and **persistent activity in left-hemisphere cognitive regions** → impaired vasocongestion and lubrication → orgasmic failure.

Concept — HSDD is defined by distress, not by desire level or hormone level: ageing lowers desire without raising HSDD, no hormone threshold predicts it, and relationship causes must be excluded first; mechanistically it is an **excitation–inhibition imbalance**, which is what the licensed drugs target.

Evaluation

- **Clinicians must initiate** — most women with low desire, **postmenopausal women even more so, never raise it.**
- **Decreased Sexual Desire Screener (DSDS): validated, five questions, self-administered;** identifies **generalised acquired HSDD** pre- and postmenopause; usable by non-specialists with good accuracy against expert interview. Because DSM-5 merges HSDD and FSAD, **also ask about genital and nongenital arousal.**

TABLE 77.14 DECREASED SEXUAL DESIRE SCREENER (CLAYTON ET AL., 2009; GOLDFISCHER ET AL., 2008)

Item	Question	Role
1	Desire previously good and satisfying?	Acquired
2	Has desire decreased?	Change
3	Bothered by it?	Distress
4	Would like it to increase?	Wish for treatment
5a–g	Contributors: surgery, depression, injury, illness; drugs or alcohol; pregnancy, childbirth, menopause; other sexual problems; partner's problems; relationship dissatisfaction; stress or fatigue	Primary vs secondary
Answers		Interpretation
No to any of 1–4		Not generalised acquired HSDD
Yes to all 1–4 and No to all 5a–g		Generalised acquired HSDD
Yes to all 1–4 and Yes to any 5a–g		Clinician judges whether another primary diagnosis applies; comorbid arousal or orgasm disorder does not exclude HSDD

- **After a positive screen:** onset, duration, **behavioural adaptation and avoidance**, level of distress.

TABLE 77.15 CONDITIONS ASSOCIATED WITH LOW SEXUAL DESIRE

Category	Conditions
Cardiometabolic	Hypertension, DM (types 1 and 2), metabolic syndrome
Endocrine and reproductive	Hypo- and hyperthyroidism, hyperprolactinaemia, polycystic ovary syndrome, postmenopausal status
Renal	Renal failure
Neurological	SCI, MS, neuromuscular disorders, Parkinson disease, dementia
Dermatological	Vulval dermatoses
Oncological	Breast cancer and treatment
Psychiatric	Depression

- **Pattern:** every desire-relevant endocrine axis appears. **Trap:** both hypo- and hyperthyroidism; renal failure included.
- **Depression and FSD are strongly bidirectional** — FSD raises depression risk even more steeply; screen every patient with HSDD for depression.

TABLE 77.16 CLASSES AND EXAMPLES OF MEDICATIONS THAT MIGHT BE ASSOCIATED WITH LOW SEXUAL DESIRE (MODIFIED FROM CLAYTON ET AL., 2018)

Class	Examples
Psychotropic	SSRIs, serotonin–noradrenaline reuptake inhibitors (SNRIs), tricyclics, antipsychotics, anxiolytics (alprazolam, diazepam)

Class	Examples
Cardiovascular	ACE inhibitors, β-blockers (atenolol, metoprolol, propranolol), calcium channel blockers , clonidine, digoxin, amiodarone, diuretics (hydrochlorothiazide, spironolactone), lipid-lowering agents
Hormonal	Antiandrogens (flutamide), gonadotrophin-releasing hormone (GnRH) agonists , oral contraceptives
Anticonvulsants	Carbamazepine, phenytoin, primidone
Analgesics	Non-steroidal anti-inflammatory drugs (NSAIDs), opiates
Illicit	Amphetamine, cocaine, heroin, marijuana
Others	Histamine receptor antagonists, alcohol , indomethacin, ketoconazole, chemotherapy

HORMONAL MECHANISMS OF DRUG-INDUCED LOW DESIRE

Mechanism	Drugs
Reduced testosterone production	Combined hormonal contraceptives , GnRH analogues, exogenous glucocorticoids
Antiandrogenic	Spironolactone , cyproterone acetate , flutamide , finasteride
Raised SHBG	Oral oestrogens , combined hormonal contraceptives , tamoxifen , thyroxine
Raised prolactin	Antipsychotics

- **Traps:** combined hormonal contraceptives act by two mechanisms; thyroxine raises SHBG; spironolactone is both diuretic and antiandrogen.
- **Disorders overlap and compound:** HSDD → poor arousal → poor orgasm → pain (penetration without lubrication); or hypo-oestrogenic dryness → dyspareunia → secondary low desire. **Temporal order indicates primary vs secondary**; establish **lifelong vs acquired**, **situational vs generalised**, prior function and relationship history.
- **Urogynaecological history:** menstrual cycle, **vasomotor symptoms**, STIs, LUTS, urinary and faecal incontinence, POP, **high-tone pelvic floor dysfunction**.

COMBINED ORAL CONTRACEPTIVES AND DESIRE

May lower desire	May raise desire
Antiandrogenic effect; reduced lubrication	Less pregnancy fear ; improved dysmenorrhoea , menorrhagia , endometriosis

- **Evidence on combined oral contraceptives is conflicting.** Menstrual irregularity → hyperprolactinaemia or hypothyroidism. Bilateral salpingo-oophorectomy before natural menopause, pelvic trauma or radiotherapy point to an anatomical cause.
- **Examination is not required for diagnosis** but reveals contributors: **atrophy**, **vestibular cotton-swab sensitivity**, **dermatoses**, **ischial spine tenderness (pudendal nerve)**, **altered anal tone or bulbocavernosus reflex latency (lumbosacral pathology)**, PFM overactivity, urethral hypermobility, positive cough stress test, POP.
- **Laboratory tests rarely help:** no biomarker confirms or excludes HSDD; prolactin or thyroid function only if signs suggest; androgens do not correlate with function and testosterone assays are unreliable at female levels.

Treatment

- **Psychosocial and biological;** sequence to address the most distressing aspect first.
- **Office-based PLISSIT counselling:** permission → limited information → specific suggestions → **intensive therapy** = referral for individual or couples therapy.

- **Correct modifiable factors first**; if HSDD persists, add **psychological therapy** and/or **central nervous system (CNS) or hormonal medication**.

FIG. 77.13 TREATMENT ALGORITHM FOR HSDD (CLAYTON ET AL., 2018)

Step	Action
1	Routinely screen — direct questioning and/or DSDS
2	Low desire with distress → sexual, medical, psychosocial history and focused examination
3	Generalised, acquired, not solely due to another condition or substance → consider pharmacotherapy: US Food and Drug Administration (FDA)-approved flibanserin; off-label transdermal testosterone, bupropion
4	Evaluate desire, distress, quality of life ; continue or adjust
Refer	Situational low desire refractory to counselling; significant psychiatric illness; unresolved sexual trauma or abuse

- The 2018 algorithm predates bremelanotide approval.

Psychological Treatment

- Detailed discussion online.

Medical Treatment

- **Flibanserin and bremelanotide are the only FDA-approved drugs for low desire, in premenopausal women only**; all others are off-label (discussed online).

FLIBANSERIN VS BREMELANOTIDE

	Flibanserin (Addyi)	Bremelanotide (Vyleesi)
Class	Nonhormonal central 5-HT _{1A} agonist and 5-HT _{2A} antagonist	Melanocortin receptor agonist, central
Net effect	Serotonin ↓; dopamine and noradrenaline ↑ — excitation up, inhibition down	Dopamine release in motivational, arousal and appetitive regions
Indication	Acquired generalised HSDD, premenopausal (first FDA-approved); not approved postmenopause	Premenopausal HSDD
Dose	100 mg orally at bedtime (continuous)	1.75 mg subcutaneous autoinjector ≥45 min before activity; ≤1 daily; ≤8 per month
Onset, response	Up to 8 weeks; roughly half respond	On demand
Efficacy	Phase III superior to placebo: desire, distress, satisfying events	Superior to placebo for desire, arousal, satisfying events; phase III FSFI desire improved; pivotal RCTs (Kingsberg et al., 2019): desire ↑, distress ↓
Adverse effects	Dizziness, somnolence, nausea, fatigue; mild-moderate; more discontinuations than placebo	Nausea, flushing, headache; hyperpigmentation (originated as 1980s sunless tanning agent)
Contraindications and restrictions	Alcohol contraindicated on label (hypotension, syncope); Risk Evaluation and Mitigation Strategy (REMS) — prescriber and pharmacy training, knowledge assessment, enrolment	Uncontrolled hypertension and/or cardiovascular disease

	Flibanserin (Addyi)	Bremelanotide (Vyleesi)
Interactions	No clinically significant interaction with ethinylestradiol–levonorgestrel	—

- **Alcohol evidence:** hypotension and syncope rare in users and non-users; **no excess adverse events in pivotal trials; post-approval interaction study showed no significant effect** — the label contraindication reflects residual concern.

Concept — *Both licensed drugs are central, not genital: flibanserin lowers the serotonergic brake, bremelanotide presses the melanocortin–dopamine accelerator* — mirroring their continuous versus on-demand use.

KEY POINTS: HYPOACTIVE SEXUAL DESIRE DISORDER

- Low desire has biological, psychological and social elements.
- The DSDS identifies generalised acquired HSDD.
- Concurrent sexual problems can exacerbate low desire.
- Systemic comorbidities and urogynaecological problems can interfere with desire.
- Flibanserin and bremelanotide are the only FDA-approved treatments for acquired generalised HSDD in premenopausal women.
- All other options, hormones included, are off-label.

FEMALE ORGASMIC DISORDER

Introduction

- **FOD:** marked delay, infrequency or absence of orgasm, or reduced intensity, for ≥ 6 months in 75–100% of sexual encounters. **Subtypes:** lifelong or acquired; generalised or situational; psychological or combined.
- **Second commonest FSD, closely behind HSDD;** prevalence **higher in Asia** than in the US, Europe and Central/South America. **Often coexists** with another dysfunction — arousal most, then lubrication, desire, pain; vaginismus rare.

DSM-IV-TR VS DSM-5 FEMALE ORGASMIC DISORDER

	DSM-IV-TR	DSM-5
Prerequisite	Despite a normal excitement phase	Removed — FOD can occur with intact arousal
Dimensions	Frequency and latency	Frequency, latency and intensity

- **Exclude before diagnosis:** medical conditions, drugs (SSRIs, other psychotropics), psychosocial causes (severe interpersonal distress, stressors), and **insufficient stimulation** — a woman whose partner's ED or premature ejaculation deprives her of stimulation cannot be diagnosed with FOD.
- **Distress is mandatory:** about half of women with inconsistent orgasm are not distressed.
- **Normal variation:** **clitorally triggered orgasm is normal;** most women reach orgasm by masturbation, **partnered orgasm is less frequent** as many need added manual stimulation — **coital anorgasmia** without clitoral stimulation is not FOD.

Assessment

- **Comprehensive sexual history** of nature and onset, with other domains (**FOD often accompanies interest and arousal problems**).

FOD SUBTYPES

Subtype	Profile	Usual cause
Lifelong generalised	Younger; never orgasmic in any context	Unfamiliarity with self-stimulation or partnered sex; poor sex education
Acquired	Previously orgasmic, now distressingly unable	Inadequate stimulation, context, physiology
Generalised / situational	All stimulation, situations, partners / specific conditions only	—
Psychological	Cognitive, affective or relational factors, no medical or substance contributor	—
Combined	Psychological plus medical or substance factors	—

- **Assess bother and distress; partnered orgasm frequency predicts overall sexual satisfaction.**

RISK FACTORS FOR ANORGASMIA

Domain	Factors
Socio-demographic	Lower socioeconomic status and education, poorer physical health
Psychological	Anxiety, depression, cognitive-affective factors, body-image misbeliefs, negative thinking, sex guilt
Relational	Conflict, dissatisfaction, poor communication or intimacy
Developmental and cultural	Restrictive attitudes; abuse or childhood maltreatment; inexperience; deficient sex education
Medical	Hypothyroidism, arthritis, hypertension, chronic pain, asthma, DM, coronary disease, malignancy (breast, colorectal, gynaecological), neuromuscular disease, SCI, MS
Drugs	SSRIs, tricyclics, antipsychotics, lithium, barbiturates, narcotics, amphetamines, digoxin, antiandrogens, chemotherapy, antihypertensives

- **Attribution is often impossible** — disease, treatment, psychological sequelae and the drug's indication (nerve damage, anxiety, depression) all impair orgasm.
- **Questionnaires with orgasm items** (FSFI, Changes in Sexual Functioning Questionnaire, BISF-W) **track response but cannot diagnose FOD.**

Psychosocial Treatment

- Detailed discussion online; options are **education (the hallmark), directed masturbation, sensate focus, coital alignment, mindfulness.**

Medical Treatment**MEDICAL TREATMENT OF FOD**

Context	Agent or device	Mechanism and evidence
With desire or arousal problems (better arousal → better orgasm; raise stimulus intensity or receptor sensitivity; cholinergic or cyclic GMP pathways)	Sexual enhancement products (vibrators, erotica)	No prescription
	EROS Therapy device	Only FDA-cleared FSD device; prescription clitoral vacuum engorging clitoris and labia; about half with arousal and satisfaction problems gained orgasm (Billups et al., 2001)
	PDE5 inhibitors	

Context	Agent or device	Mechanism and evidence
		Block cGMP breakdown → vulvovaginal vasodilatation; sildenafil improved arousal and orgasm in a double-blind RCT (Caruso et al., 2003) and SSRI-associated orgasmic dysfunction (Nurnberg et al., 2008)
	Systemic testosterone ± oestrogen; tibolone (oestrogenic, progestogenic, androgenic)	Improve desire, arousal, orgasm in postmenopausal women with low testosterone or oestrogen — central (limbic) and vulvovaginal effects; transdermal testosterone raised desire, orgasm and satisfaction (Reis and Abdo, 2014)
	Local or systemic oestrogen	Epithelial cellularity, genital flow, lubrication ↑ — all four domains; vaginal oestrogen helps orgasm indirectly via dyspareunia and arousal
Subjectively sufficient arousal	Enhancement products	Increase stimulation intensity and frequency
	Oxytocin, on demand	Short-acting; synergises with sex hormones to facilitate orgasmic contraction; hypothalamic origin, released in arousal and orgasm

- **Hormonal data for FOD as the primary complaint are sparse — mostly secondary outcomes of studies of other domains.**

Medication-Induced Female Orgasmic Disorder

- **SSRIs commonly impair desire, arousal and orgasm.** Sequence: **establish onset** (depression itself causes dysfunction; treatment may help over weeks) → if a side effect occurs while depression responds, **wait for tolerance** (weeks to about 4 months) → **persistent beyond 3–6 months: reduce the dose or switch — bupropion or mirtazapine (or an SNRI).**

Surgical Treatment

- **Reversible structural causes such as clitoral phimosis may be treated surgically.**

KEY POINTS: FEMALE ORGASMIC DISORDER

- **Exclude medical, pharmacological and psychosocial causes before diagnosis.**
- **A comprehensive history of nature and onset is essential.**
- **FOD often accompanies interest and arousal problems.**
- **Education is the hallmark of all FOD treatment.**
- **Psychosocial options: directed masturbation, sensate focus, coital alignment, mindfulness.**
- **Hormonal treatment for FOD needs further safety and efficacy research.**

Concept — FOD is a *diagnosis of exclusion anchored in distress* — inadequate stimulation, partner dysfunction, drugs and disease must go first, and coital anorgasmia alone is normal variation — so treatment splits on whether **arousal is intact: if not, treat arousal; if so, increase stimulation.**

FEMALE SEXUAL AROUSAL DISORDER

- **ICSM 2017 used DSM-IV-TR categories rather than DSM-5, consistent with the ICSM Definitions committee and ISSWSH nomenclature: separating desire and arousal eases assessment and treatment, whereas FSIAD is new and controversial.** Overlap is common, but **treatment targets the primary disorder.**

Introduction

- **ICSM definition:** persistent or recurrent **inability to attain or maintain arousal until completion**, on **adequate subjective assessment of the genital response**. **DSM-5 merged FSAD with HSDD as FSIAD**; neither DSM matches women's experience, and **no expert consensus on FSIAD** exists.
- **Arousal described since the 1970s** and debated because of **discordance**:

SUBJECTIVE VS GENITAL AROUSAL

Subjective sexual arousal	Genital arousal
The woman's perception of her genital response	Physiological activation — lubrication, vasocongestion
Self-report	Physiological measurement

- **The two correlate weakly**; components of arousal may be **processed differently in a woman's appraisal**.

Interview Assessment of Sexual Arousal

- **The individual's subjective experience is crucial**; a **semistructured interview** defines her concept of arousal — erotic images, excitement, pleasurable genital sensation, noticed lubrication.

TABLE 77.17 EXAMPLES OF HELPFUL QUESTIONS AS SUGGESTED BY THE FEMALE SEXUAL DYSFUNCTION–MEDICAL AND PSYCHOLOGICAL TREATMENTS COMMITTEE (KINGSBERG AND ALTHOF, 2009)

Theme	Question	Detects
Frequency	How often is she sexually excited?	Baseline
Triggers	Which situations excite her (attractive person, caressing, images, stories)?	Stimulus preference; women reported more responsive to narrative, men to visual
Fantasy	Fantasies, and arousing scenarios?	Cognitive arousal
Partner	How easy is arousal with the current partner?	Partner-specific problem → relationship issue
Genital sensation	Warmth, tingling, pleasure, wetness?	Genital awareness
Dreams	Erotic dreams?	Preserved capacity
Beliefs	Is her excitement healthy?	Guilt (double-standard upbringing), childhood sexual trauma
Change	What interferes now, if arousal was easier before?	Acquired cause

- **Trap:** a **partner-specific answer redirects the diagnosis to relationship difficulty**.

Potential Treatments for the Psychosocial Aspects of FSAD

- **Careful evaluation is mandatory** (ICSM). Investigate at least one of: **(1) cultural or religious guilt and inhibition**; **(2) partner-specific problems**; **(3) distracting cognitions** (attractiveness, performance, trauma memories); **(4) high stress or poor coping**; **(5) negative focus** (genetic disposition, attachment); **(6) over 40: no partner, or partner's illness or ED**.

PSYCHOSOCIAL TARGETS AND INTERVENTIONS

Target	Intervention
Guilt	No established treatment ; online role-play may identify a clergy member or community leader with a moderate stance on pleasure , possibly more effective than an outside professional (Christensen et al., 2013)
Distraction	Mindfulness within sex cognitive behavioural therapy (CBT) — focus on own sensations, notice and redirect negative thoughts

Target	Intervention
Stress	Mindfulness meditation (scarce data)
Body image	Weight loss, nutrition, exercise
Mood and attachment	Depression and anxiety are poor prognostic factors for sex therapy; treating them may help

PHARMACOTHERAPY FOR FEMALE SEXUAL AROUSAL DISORDER

FSAD PHARMACOTHERAPY

Agent	Class and mechanism	Evidence	Verdict
Testosterone (transdermal patch)	Direct action on vagina and genital structures, independent of oestrogen	Desire and arousal — Level 1; oral testosterone undecanoate enhanced genital arousal in hypogonadotrophic hypogonadal women (Tuiten et al., 2000)	Best evidenced
Tibolone	Selective tissue oestrogenic activity regulator (STEAR); metabolites act differently by organ	Mood and libido improved postmenopause	Further research needed
PDE5 inhibitors	Vasoactive, by analogy with ED	Contradictory RCTs, lacking efficacy; Pfizer ended female sildenafil programme 2004	Level 2 only in neurovascular disease: type 1 DM, SCI, MS, SSRI-induced FSAD
Alprostadil (prostaglandin E1), topical	Vasodilator; enhances sensory afferents	No consistent benefit (Kielbasa et al., 2006)	Not supported
L-arginine + yohimbine	NO precursor + adrenergic antagonist	Physiological arousal ↑ (vaginal photoplethysmography), subjective arousal unchanged (Meston and Worcel, 2002)	Demonstrates discordance
Apomorphine, sublingual	Nonselective D1/D2 dopamine agonist, central	Improved premenopausal FSAD (Caruso et al., 2004); with vibrator: arousal and lubrication ↑, orgasm unchanged (Bechara et al., 2004)	Small studies
Bupropion	Dopamine–noradrenaline reuptake inhibitor, no serotonergic effect	Pro-sexual across all domains; Cochrane (Taylor et al., 2013): 150 mg twice daily improves SSRI-induced dysfunction — Level 2	Choice for SSRI-induced dysfunction
Oxytocin	Parturition and lactation neuropeptide; prosocial behaviour	Potential arousal and orgasm agent	Investigational

Concept — Because **subjective and genital arousal are discordant**, peripheral vasoactive drugs can raise measured flow without the woman feeling aroused; the agents with benefit act **centrally** (testosterone, dopaminergics, bupropion) or **restore a damaged neurovascular substrate**.

Female Persistent Genital Arousal Disorder

- ICSM: spontaneous, intrusive, unwanted genital arousal without interest or desire, usually unpleasant, **not relieved by at least one orgasm**, lasting hours to days.

TABLE 77.18 LEIBLUM AND NATHAN CRITERIA FOR PGAD IN FEMALES (2001)

No.	Criterion
1	Physiological arousal (genital and breast vasocongestion and sensitivity) persists hours to days and does not remit spontaneously
2	Does not remit with ordinary orgasm
3	Occurs without subjective desire or arousal
4	Triggered by sexual stimuli, nonsexual cues or nothing
5	Intrusive and unwanted, causing at least some distress

- **Pattern:** each criterion separates PGAD from normal arousal. **Trap:** genital arousal without desire — the reverse of HSDD.

PGAD DIFFERENTIAL DIAGNOSIS

PGAD	Hypersexuality	Clitoral priapism
Unwanted arousal without desire	Out-of-control thoughts and behaviour with desire	Painful clitoral engorgement

Psychosocial Characterisation and Aetiological Factors

- **Profile** (limited evidence): negative appraisal of genital sensations, more depression and panic attacks, sexual victimisation, sexually conservative thinking with maladaptive cognition during sex; FSFI shows no dysfunction but an unsatisfying sex life.
- **Aetiology** (poor data): periclitoral mass, pelvic varices, Tarlov cysts; venlafaxine use or SSRI discontinuation; biological versus psychological proportions unknown.

Treatment Targets**PGAD TREATMENTS**

Treatment	Evidence
Duloxetine, pregabalin, varenicline	Case series
Electroconvulsive therapy	Small studies
CBT — anxiety management, response prevention, dyadic issues	Improves sexual and emotional symptoms
Endoscopic surgery for lumbosacral annular tear causing sacral radiculopathy	Majority success (Kim et al., 2023)

KEY POINTS: FEMALE SEXUAL AROUSAL DISORDER

- FSAD: persistent or recurrent inability to attain or maintain arousal until completion.
- DSM-5 merges FSAD with HSDD as FSIAD.
- Defining the patient's concept of arousal is the first assessment step.
- Testosterone acts directly on genital structures and can improve desire and arousal.
- PDE5 inhibitors may be considered when arousal disorder stems from conditions affecting genital neurovascular supply.
- Use the Leiblum criteria to define PGAD.

GENITOPELVIC PAIN AND PENETRATION DISORDER

Pain and Sex

- **Sexual pain is common, affects partners, and is neglected, poorly understood and often mismanaged;** it may arise from **vulva, vagina, cervix, uterus, adnexa, PFMs or their nerves.**

SEXUAL PAIN TERMINOLOGY

Term	Definition
Dyspareunia	Pain during intercourse
Vaginismus	Involuntary vaginal muscle contraction interfering with intercourse
Vulvodynia	Vulval discomfort, typically burning, without relevant visible findings (Moyal-Barracco and Lynch, 2004)
Localised / generalised	Confined (vestibule, entrance) / whole vulva
Provoked / unprovoked	Triggered by external stimulation / spontaneous

- **Commonest vulvodynia: localised to the vestibule and provoked by intercourse** — termed **provoked vestibulodynia (PVD)** by the **International Society for the Study of Vulvovaginal Disease (ISSVD)**; affects **about 10–12% of women**; commonly coexists with **irritable bowel syndrome (IBS)** and **fibromyalgia**.

Provoked Vestibulodynia

PRIMARY VS SECONDARY PVD

	Primary	Secondary
Onset	From the first penetration of any kind (tampon or sexual)	After pain-free penetration ; acquired, late
Course	Early onset, longer duration	—
Associations	Nulliparity; positive family history; enuresis, dysmenorrhoea	—
Signs	Local inflammation less common	Larger vestibular area; clitoral hood most painful
Frequency	Less common; commoner in Hispanic women	Commoner (~two-thirds)

Pathogenesis

- **Multifactorial, incompletely understood;** detailed discussion online.

Clitoris and Clitorodysnia

- **Vestibulodynia may present from the clitoris; clitoral pain is greatly underdiagnosed and undertreated. Clitoral examination is neglected** — not taught in gynaecology, and **urology teaches penis and foreskin examination but not the clitoris**, though **hood and foreskin share function and pathology**.
- **Mechanism: clitoral adhesions** → poor drainage of keratin → **smegma and squamous accumulation** → **smegmatic pseudocysts, keratin pearls** → irritation; **erythema and infection** → **clitorodysnia**. Adhesions also predispose to **yeast infection, urinary tract infection (UTI), genital trauma and lichen sclerosus**; a **substantial minority** of women at vulvoscopy have some adhesion.

CLINICAL PRESENTATION

Menopause and Dyspareunia

- **Sexual activity continues into old age.** Vulvovaginal atrophy needing treatment affects a substantial minority of postmenopausal women, and **many with atrophy report dyspareunia**.
- **Symptoms:** dryness, burning, poor lubrication, pain, impaired function; **urgency, dysuria, recurrent UTI**. **Current term: genitourinary syndrome of menopause (GSM).**
- **Menopausal dyspareunia is often vestibular** — atrophy more marked in the vestibule than the vagina.

Previous Genitourinary Surgery

- **Synthetic mesh in POP surgery is under scrutiny for life-changing complications:** erosion, pain, infection, bleeding, **dyspareunia**, organ perforation, urinary problems — **possibly years later; most RCTs lack medium- and long-term safety data**. Detailed discussion online and in Chapter 119.

Figure legend

- **Fig. 77.14** (Aerts et al., 2018): **normal glans without adhesions**, prepuce retracted to the balanopreputial sulcus, full corona visible — **the glans clitoris should be examined routinely**.

TREATMENT

TREATMENT OF GENITO-PELVIC PAIN AND PVD

Modality	Content	Mechanism and evidence
Education	Vulval hygiene, avoiding irritants, behaviour modification, stress reduction, pain education	Pain improved in the majority over 6–36 months (Fowler, 2000)
Biofeedback	Visual electromyography (EMG) feedback	PFM tone has active (contractile) and passive (viscoelastic) components; many asymptomatic women cannot contract correctly on verbal instruction; Glazer protocol (1995): about half pain-free at 16 weeks
Dilators	Graduated insertion	Control fear, promote relaxation; optimal duration unknown
Electrical stimulation	Domiciliary, or in-clinic + home PFM exercises	Proprioception and circulation ↑, nociception ↓, endorphins ↑ ; improved dyspareunia and function
Manual therapy	Stretching, massage, myofascial techniques	Cornerstone of physical therapy: relaxation, trigger-point release, circulation, posture, introital widening, desensitisation
Multimodal physical therapy	Manual + stretching + PFM control ± biofeedback + education	Current standard in PVD
Topical therapy	North American Menopause Society (NAMS): nonhormonal lubricants and moisturisers with regular sexual activity first-line (often inadequate); oestrogen	Oestrogen deficiency raises vulval sympathetic, parasympathetic and sensory fibre density; oestrogen lowers it; topical > systemic; vestibular > vaginal
Surgery — vestibulectomy	Excision of the vestibule	More improve than with nonsurgical or no treatment (Goetsch, 2007)
Multidisciplinary, multimodal care	Combined medical, physical, psychological	Addresses pain with its psychosexual and relational consequences

Figure legend

- **Fig. 77.15: complete vestibulectomy for neuroproliferative vestibulodynia, removing all vestibular tissue to within 1–2 mm of the urethral meatus.**

KEY POINTS: GENITOPELVIC PAIN AND PENETRATION DISORDER

- Sexual pain may arise from any genital organ, the PFM's or their nerves.
- About 10% of women have PVD, often with IBS and fibromyalgia.
- Postmenopausal atrophy causes dyspareunia.
- Biofeedback helps vulvodynia, especially with hypertonic pelvic floor overactivity.
- Multimodal physical therapy is the mainstay for PVD.
- Chronic vulval pain needs integrated multidisciplinary care.

Concept — PVD is a **peripheral pain problem amplified by a hypertonic pelvic floor and nerve**

proliferation: physical therapy relaxes the muscle, vestibular oestrogen reduces nerve density, vestibulectomy removes the sensitised tissue — the more localised and neuroproliferative the disease, the stronger the case for excision.

PELVIC SURGERY AND FEMALE SEXUAL DYSFUNCTION

Introduction

- Major gynaecological, urological and colorectal surgery commonly causes FSD; radical cystectomy (oncological or benign, e.g. interstitial cystitis) is **the major urological cause**. Mesh is under scrutiny, yet **only half of trials measure dyspareunia**; female outcomes are less studied than male.

PELVIC OPERATIONS AND FEMALE SEXUAL FUNCTION

Operation	Consequences	Determinants	Notes
POP repair	Restores anatomy; de novo dyspareunia with any technique ; FSFI domains, especially lubrication , may worsen	Preoperative body image , approach, mesh	See below
Radical cystectomy	Desire loss > orgasm disorder > dyspareunia > reduced lubrication ; about half can have intercourse; half less satisfied	Nerve, clitoral vascular, vaginal factors	Partner relations worsen in a substantial minority; a quarter stop intercourse
Rectal surgery	Benign (proctocolectomy for inflammatory bowel disease, IBD): activity may increase; malignant: more dysfunction (pelvic plexus injury)	Operation, tumour level, neoadjuvant treatment	Peaks at 1 month; lower with transanal approach

Hysterectomy

- **Commonest pelvic operation in women**, for benign or malignant indications; **sexual function is often not assessed afterwards**.
- **Subtotal hysterectomy**: most unchanged or improved, **a minority deteriorate**, attributed to **vaginal sensory loss and the indication for surgery**; **clitoral thermal and vibratory sensation unchanged**.
- **Radical hysterectomy at 6 months**: **lubrication distress and dyspareunia commonest**; most women active before the cancer diagnosis are active again at 12 months, at **lower frequency**. Ovarian cancer surgery gives similar rates.
- **Abdominal versus vaginal approach for simple hysterectomy makes no difference** to postoperative sexual function.

Pelvic Organ Prolapse Repair

- **Concomitant hysterectomy ± cervical resection is controversial.** Native-tissue vaginal repair: function improves in about half, unchanged in over a third, worse in a minority.
- **Transvaginal mesh gives sexual function comparable to native tissue** (two systematic reviews); **vaginal length does not affect sexuality**; **persistent dyspareunia is linked to anterior vaginal mesh versus laparoscopic sacrocolpopexy**; **dyspareunia higher after vaginal than abdominal repair.**

Cystectomy

- **Sexuality-determining factors:** (1) preserve nerves along the lateral vaginal walls; (2) avoid clitoral devascularisation by preserving a distal urethral stump; (3) vaginal reconstruction may alter canal dimensions.
- **Sexuality-preserving technique:** urethral transection at the bladder neck (clitoral vasculature) + retrograde dissection of bladder off vagina (neurovascular bundle). **Reconstruction orientation affects vaginal tightness.** Diversion type does not change sexual outcome.
- **Genital-sparing cystectomy, when oncologically possible:** earlier return to sexual activity (~6 vs 14 weeks) and better function.
- **Counselling gap:** far more patients want information than ask — **the onus is on the clinician**; a third of cancer patients wished for counselling and sex-therapy or partner referral; after benign simple cystectomy, most were **not counselled preoperatively** and **almost none asked about sex at follow-up.**

Rectal Surgery

LOW ANTERIOR VS ABDOMINOPERINEAL RESECTION

Low anterior resection	Abdominoperineal resection
Dysfunction in a substantial proportion	Dysfunction in nearly all

- **Independent risk factors:** preoperative chemotherapy, tumour below the peritoneal reflection, abdominoperineal and intersphincteric resection, Hartmann's procedure; include in preoperative counselling.

Genital Cosmetic Surgery and Female Sexual Dysfunction

- **Procedures:** labiaplasty, clitoral hood reduction, hymenoplasty, perineoplasty, labia majora augmentation, vaginoplasty, **G-spot amplification.** **Rising interest, limited evidence** — no standard nomenclature, few functional, psychosexual or complication data.
- **Counsel:** lack of high-quality efficacy data; complications — pain, bleeding, infection, scarring, adhesions, altered sensation, dyspareunia, reoperation.
- **Pathway:** screen for body dysmorphic disorder (BDD) → refer if psychological concerns → surgeon discloses experience, outcomes, complication rates. Motives: cosmesis, self-esteem, sexual function; wanting alteration does not mean she is abnormal.
- **Surgeons need sexual-medicine training to withhold surgery in sexual dysfunction, mental impairment or BDD,** and surgical expertise to deliver satisfaction.

KEY POINTS: PELVIC SURGERY AND FEMALE SEXUAL DYSFUNCTION

- Patients' wish for counselling exceeds what they receive.
- Pre- and postoperative counselling on sexual function is necessary.

- **Sexuality-preserving technique may be used, but never at the expense of oncological success.**

Concept — Pelvic surgery damages female sexual function through the **same three substrates as in the male** — **nerves (lateral vaginal wall, pelvic plexus), clitoral vasculature (distal urethra) and genital geometry (vaginal length and calibre)**; preservation technique protects each within oncological limits, and the dominant failure is communicative — patients are not asked.

LOWER URINARY TRACT DYSFUNCTION AND FEMALE SEXUAL DYSFUNCTION

Introduction

- **LUTS**: the **subjective indicator of disease or change as perceived by patient, carer or partner**. **International Consultation on Incontinence–Research Society (ICI-RS) 2017 think tank** addressed LUTS and sexual function. **5th International Consultation on Incontinence (2012)**: LUTS in **nearly half of adults ≥20 years**, OAB about one in ten, urinary incontinence (UI) fewer, severe stress urinary incontinence (SUI) a small minority.

Sexual Lifestyle and Communication

- **Sexual health**: physical, emotional, mental and social well-being in relation to sexuality, **not merely absence of disease**. **Three National Surveys of Sexual Attitudes and Lifestyles (Natsal)** show major change; in women **age reduces frequency and range**.

TABLE 77.19 CHANGE IN SEXUAL PRACTICES, 1990 TO 2010 (MODIFIED FROM RANTELL ET AL., 2017)

Behaviour	Direction
Lifetime partners	About doubled
Sexual occasions per month	Fell
Vaginal sex, past month	Fell slightly
Oral sex, past month	Rose
Anal sex, past year	More than doubled
Same-sex experience	About fourfold
Masturbation, past 4 weeks	About a third (2010 only)

- **Pattern**: more partners and diversity, less frequency. **Trap**: **anal sex and same-sex experience rose most** — history-taking must not assume penile–vaginal intercourse.
- **Communication is problematic**: **patients feel relief when asked**; most gynaecologists ask about activity but **fewer than half ask about problems**; **women may prefer a nurse**. **Clinician barriers**: fear of opening an unmanageable problem, lack of time, resources and training, doubts about knowledge, fear of offence, discomfort, low awareness, **gender, race or age discordance**.

Epidemiology

- **About 40% of women have a lifetime sexual problem, but only ~8–12% a distressing, diagnosable disorder** (IUGA and International Continence Society (ICS) joint report; Shifren et al., 2008).
- **Pelvic floor disorders (PFDs) and sexual function**: debated (heterogeneity), but **generally harmful** — dyspareunia, low arousal, infrequent orgasm; **up to nearly half of women with UI/LUTS have sexual dysfunction, pain and HSDD most frequent**; function tracks **body image and POP bother**, especially stage ≥2, in desire and satisfaction; **chronic pelvic pain (CPP)** causes dyspareunia and **avoidance**.
- **Pelvic floor strength correlates with better function**; **high tone with dysfunction**. Treating PFDs often improves sexual function.

Aetiology and Classification

- **Systemic disease** (endocrine, neurological, cardiovascular, dermatological, psychiatric, surgical, cancer) predisposes to FSD via altered signalling, endocrine milieu, neurotransmitters, tissue damage, vascular and neural change.

IUGA/ICS FEMALE SEXUAL DYSFUNCTION SYMPTOMS

Group	Symptoms
Lower urinary tract	Coital UI; orgasmic UI; penetration UI (penile, manual, device); coital urgency; postcoital LUTS (frequency, urgency, dysuria, suprapubic tenderness); receptive urethral intercourse
Prolapse-specific	Abstinence due to POP; vaginal wind; vaginal laxity; obstructed intercourse (prolapse, short vagina, lichen planus or sclerosis)
Pain	Dyspareunia — superficial (introital) vs deep (mid/upper vagina); penile-movement dyspareunia; vaginismus; vaginal dryness; hypertonic PFM (contractile activity and/or passive stiffness); noncoital sexual pain; postcoital pain; vulvodynia — ≥3 months without identifiable cause

- **Trap: vulvodynia needs ≥3 months; superficial vs deep dyspareunia is defined by depth of penetration.**

Clinical Signs and Investigations

- Detailed discussion online.

Lower Urinary Tract Symptoms and Female Sexual Dysfunction: Links and Treatments

Urinary Incontinence and Overactive Bladder

- **A quarter to a half of women with UI have FSD; all UI forms affect all phases and domains.** Burden: concealment strategies, apathy, guilt, **markedly more major depression**, threatened femininity, shame, isolation. **Main drivers: low libido and self-esteem from fear of leakage; severe UI roughly doubles odds of low libido, dryness and dyspareunia.**
- **OAB is contradictory: EpiLUTS shows association; in a large treatment study only a minority reported sexual impact, and OAB did not predict loss of interest in women (unlike men); tolterodine (immediate and extended release) improved desire, arousal, lubrication, orgasm and satisfaction for up to 6 months.**
- **Pairings: HSDD with SUI; orgasm problems with OAB and urgency urinary incontinence (UUI); SUI or OAB → poorer quality of life (King's Health Questionnaire), satisfaction and marital relations.**
- **Severity of FSD by LUTS type: bladder pain syndrome (BPS) > urodynamic detrusor overactivity (DO) > UUI and mixed urinary incontinence (MUI) > SUI > dry OAB > voiding LUTS.**
- **Mechanisms: fear of leakage, coital UI at orgasm, interrupting intercourse to void, postcoital urgency, dyspareunia, PFD.**

COITAL INCONTINENCE: PENETRATION VS ORGASM

Leakage at penetration	Leakage at orgasm
Commonest pattern	Less common; some leak at both
SUI mechanism — displacement of anterior vaginal wall and bladder neck, or raised abdominal pressure	Simultaneous detrusor contraction and urethral relaxation on urodynamics
SUI present in most	SUI present in most

- **About one in ten women with UI leak during intercourse; a questionnaire detects it.**

SUI surgery

- **Thought to improve sexual function; data limited and conflicting.** Meta-analysis: PISQ-12 slightly favoured transobturator over retropubic sling; FSFI showed no difference — **overall uncertain.**

- **Slings can harm function** (neurovascular injury, **de novo dyspareunia**); **dyspareunia is a severe indication for removing a posterior intravaginal sling.**

TVT VS TOT NEURAL EFFECTS (BEKKER ET AL., 2012)

Tension-free vaginal tape (TVT, retropubic)	Transobturator tape (TOT)
Disrupts vaginal wall autonomic innervation → altered lubrication–swelling	Spare the dorsal nerve of the clitoris

- **After SUI surgery: mostly unchanged, about a third improved, a minority worse. Preoperative coital incontinence predicts improvement; cure of coital incontinence (achieved in most surgical cases) is a major predictor**, with gains in coital pain and partner aspects.
- **ICI-RS: improvement after treating OAB/UI/SUI implies lower urinary tract dysfunction (LUTD) acts indirectly** — through leakage, urgency interruptions, pain, embarrassment, fear, self-image.

Neuromodulation

- **Sacral neuromodulation (SNM) improves LUTS and sexual function whatever the cause** — implying shared neural and vascular change; **puddendal function is abnormal in refractory OAB and non-obstructive retention and trends to improvement after SNM. Epidural spinal cord stimulation** produced pleasurable genital sensation, more activity, lubrication and orgasm; neurostimulation raises **vaginal pulse amplitude**. Whether the **male LUTS–ED four-theory model** applies to women is unresolved.

Pelvic Organ Prolapse

- **Common in gynaecology attenders; SUI, urgency, frequency, UUI each in a substantial minority;** symptoms subjective; associations with LUTS and sexual function **contradictory**.

POP AND SEXUAL FUNCTION — STUDY SERIES

Study	Measure	Verdict
Cetinkaya et al. (2013)	PISQ-12 vs stage	No correlation
Athanasίου et al. (2012)	POP vs controls	Worse, not graded with stage
Handa et al. (2004)	Complaints	No association
Weber et al. (1995)	POP/UI vs controls	Similar
Novi et al. (2005)	PISQ	Worse: satisfaction, actual frequency, orgasm with masturbation ; no difference : desired frequency, initiation, anorgasmia, partner satisfaction
Digesu et al. (2005)	Mechanism	Uterine displacement — cervix obstructs penetration

- **Pattern: POP may worsen function, but stage does not predict it;** bother and body image matter more.
- **Native-tissue repair (anterior and posterior colporrhaphy, uterosacral vault suspension, sacrospinous fixation): most improved or unchanged;** meta-analysis (Jha and Gray) — function improved and women were **several-fold more likely to have improved or unchanged dyspareunia than worse.**

POP REPAIR OPTIONS AND SEXUAL OUTCOMES

Repair	Dyspareunia	Recurrence	Status
Anterior native vs biological graft	No difference	Lower with graft	—
	No difference	—	—

Repair	Dyspareunia	Recurrence	Status
Posterior native vs biological graft			
Biological graft vs polypropylene	—	Biological higher	Biological out of favour
Polypropylene mesh	Transvaginal mesh dyspareunia under one in ten, similar for synthetic and biological (Society of Gynecologic Surgeons review)	Durable	Favoured for durability
Sacrocolpopexy	Unchanged or improved	—	Gold standard for apical prolapse

- Critical lack of sexual data after mesh repair. Combined anterior and posterior repair consistently increases dyspareunia — consider two-stage repair in sexually active women. Principles: maintain vaginal length, avoid excessive excision, preserve introital calibre.

Bladder Pain Syndrome and Interstitial Cystitis

- Sexual dysfunction is common in BPS/interstitial cystitis (IC) and lowers quality of life. Bladder wall inflammation and neuropathic pain → dyspareunia and fear → resistance to penetration → PFM overactivity, vulvodynia, vaginismus.
- BPS has the greatest sexual impact of all LUTD, mainly through pain. CPP associates with HSDD, arousal, orgasmic and pain disorders.

Hysterectomy

- Detailed discussion online.

SELF-IMAGE/BODY IMAGE CONNECTED TO FEMALE SEXUAL DYSFUNCTION AND LOWER URINARY TRACT SYMPTOMS

- Detailed discussion online.

KEY POINTS: LOWER URINARY TRACT DYSFUNCTION AND FEMALE SEXUAL DYSFUNCTION

- Almost half of women with UI and/or LUTS have sexual dysfunction.
- Sexual symptoms coexist with UI, POP and pelvic pain.
- UI harms sexual function through libido and self-esteem.
- Midurethral slings can cause FSD through neurovascular injury or de novo dyspareunia.
- Treating UI medically or surgically can improve many sexual domains.
- Sexually active women with POP may have lower satisfaction, less frequent intercourse and difficulty reaching orgasm by masturbation.

Concept — LUTD affects female sexual function **mostly indirectly** — fear, leakage, pain, self-image — which is why curing coital incontinence predicts sexual recovery; where the damage is **direct** (TVT denervation, pelvic floor hypertonicity, pudendal dysfunction reversed by SNM) it points to a **shared neurovascular substrate**, as with male LUTS and ED.

FEMALE GENITAL MUTILATION

- Detailed discussion online.

CONCLUSIONS

- **Female sexual function and dysfunction are integral to urological practice;** urologists must recognise the **bidirectional relationship** between urological disease and sexual issues. **Appropriate treatment or referral improves satisfaction and compliance.**